ASME
SETTING THE STANDARD

TEXAS A&M UNIVERSITY-
TECHNOLOGY
STUDENT SECTION

Spring 2026

HAND ERC: Dexterous Design Challenge

Rules and Guidebook

Last updated: 02/15

TEXAS A&M UNIVERSITY Engineering
MXET Mechatronics | STEM Education
COBOTICS Collaborative Robots & Intelligent Control Systems Laboratory
NSF ERC
HAND

Table Of Contents:

Introduction ----------3
Tasks and Deliverables ---------- 3
Important Dates and Timelines ---------- 4
Communication ---------- 5
Kits ---------- 6
Design Process and Check-In---------- 8
Rules ---------- 8
Resources ---------- 12
Competition Guide ---------- 14

Introduction:

Welcome to the inaugural HAND ERC Dexterous Design Challenge! This is a hands-on engineering competition that tasks teams to design, build, and program a functional tendon-driven robotic hand using a standardized kit and a defined set of constraints. This guidebook outlines the rules, expectations, timeline, and resources necessary to participate in the competition. Participants are encouraged to familiarize themselves with this document early and refer to it throughout the competition. Additional details and clarifications will be communicated by the competition coordinators as needed. If any errors or unclear wording is found in this document, please contact the coordinators over the Discord, or at AsmeTechTamu@gmail.com.

Tasks and Deliverables:

Design Deliverables (no submission required):

- Mechanical Hand
- Mechanical Wrist
- Actuation Mechanisms
- Tendon Routing Mechanisms
- Control Program
- User Interface
- Design Documentation (optional but highly recommended)
- Sensor Integration (optional)

Organizational Deliverables (submissions required):

- Initial Planner
- 5 Week Planner
- CAD files for Check-In
- Design Presentation (During competition period, more details TBA)
- Additional Materials Form

Important Dates:

02/12:
Informational. This is an opportunity to go over competition expectations, initial design decisions, as well as pick up available kit components. At least one team member from each team must attend, but it is encouraged for the entire team to attend if available.
Location: Zach 244, 7-8:30pm, (02/12).
Final day to request team name changes.

02/19(Date subject to change):
Workshop I. This workshop is an introduction to the mechanical components of tendon routing. It will cover how forces in tendon routed mechanisms work, and different degree of freedom configurations such as n, n+1, and 2n.

02/26(Date subject to change):
Workshop II. This first half of this workshop is an introduction to the software and electronics used in the kits. The second half of this workshop is an introduction to Onshape, a cloud based CAD software. For individuals who have not used microcontrollers, I2C, servo libraries, or CAD, this serves as an entry point to working with them. Individuals may attend either half, or both halves of this workshop.

03/05(Date subject to change):
Workshop III. This workshop will be used to fill any particular knowledge gap for teams. A poll will be conducted to determine the focus of this workshop.

03/20:
Deadline to submit files and plans in preparation for the design check-in. More details will be provided about what to submit closer to the check-in.

Week of 03/22:
Design check-in. This is an opportunity for teams to meet with the competition coordinators to present their progress and ensure that they are on the right track to complete their design in time. The coordinators will work with each team to make a plan for the remaining weeks of the competition.

04/11-4/12 (Exact dates subject to change):
Competition days.
Day 1 (04/11):
Design presentation. Teams will provide a brief presentation on their design process to a judging panel. Teams will be scored on presentation, design process, documentation, and critical thinking. This will compose part of the team's overall competition score. More details will be released on scoring closer to the competition date, but example scores can be seen in the Competition Guide section.
Day 2 (04/12):
Testing day. Designs will be put through a series of tests, and scored for performance on each. Cumulative score will be added to presentation score for total score. More details on competition tasks and scoring will be released close to the competition date. See the Competition Guide section for example tasks and scoring.

04/18:
Deadline to return loaned materials to the MXET program if not purchased. (See Kit section for more details)

Communication:

Major updates and communications will be sent through the ASME-Tech email, or directly from Niamh Coles or Zavier Dollaway. For more frequent updates, join the ASME-Tech Discord using the following link: https://discord.gg/6E5aHtbxCq.

If you need to contact the competition coordinators for any reason, please contact us at AsmeTechTamu@gmail.com, or directly message us through your team's private chat in the Discord. **For all emails, please be sure to CC all team members in your initial email, and include the team name in the subject line.**

For more in depth discussions, please arrange a meeting with the coordinators by contacting AsmeTechTamu@gmail.com.

Kits:

Teams will be provided with **two** sets of supplies to complete this competition. The components in these kits must be used in accordance with the design rules listed in the Rules section.

The first set of supplies is the purchasable kit. Teams are *required* to purchase one of these kits to stay registered for the competition. Teams will keep these supplies after the competition is completed.

The second set of supplies is loaned from the MXET program. This set of supplies *must* be returned to the MXET program by one week after the competition (04/18). These components are loaned to minimize kit costs for the teams. Near the end of the challenge, teams will have the option to purchase this kit to keep their design complete. This will be an additional cost, likely around $50, but is not mandatory. If loaned materials are significantly damaged, the coordinators reserve the right to ask teams to replace or reimburse the cost of the damaged materials.

The purchasable kits are $30 per team. These kits will be picked up from the . Teams are required to use the components provided in the kits, but may use additional components as detailed in the Rules section.

Purchased Materials:

Quantity	Item	Documentation Link
1	Nano V3.0 Microcontroller	https://robotics.ee.uwa.edu.au/nano/doc/nano-user-manual.pdf
1	Nano Board USB-B Cable	--
1	4x AA Battery Holder	--
1	Assorted Length M2 Bolt Set	--
25yd.	Spiderwire Braided Cord (20lb)	Amazon Page
8	M4 Bolt Set	--

20	Male to Female Wire	--
1	Forearm Base	https://drive.google.com/file/d/1lcBwfRs-omkCxLn3hqJwVsdmtAdIdQAO/view?usp=sharing
2	Forearm Cover	https://drive.google.com/file/d/1lcBwfRs-omkCxLn3hqJwVsdmtAdIdQAO/view?usp=sharing

Loaned Materials:

Quantity	Item	Documentation Link
10	MG90 Servo	https://www.electronicoscaldas.com/datasheet/MG90S_Tower-Pro.pdf?srsltid=AfmBOoo5ZVVdlZ0wfqoKlZcolpbnVhTwvvJoSpptEcWPUgIOGgFsr71r
1	MG996R Servo Motor	https://www.handsontec.com/dataspecs/motor_fan/MG996R.pdf
1	Pre-SolderedPCA9685 Servo Driver	https://cdn-learn.adafruit.com/downloads/pdf/16-channel-pwm-servo-driver.pdf
1	4xBattery Charger with Batteries	--
1	5gal. Storage Box	--

Design Process and Check-In:

During the introductory meeting and the first week, teams will design their initial Gantt chart. It is recommended that teams keep this plan up to date, and have weekly team meetings with the entire team to discuss progress, as this plan will likely change over time. Halfway through the design period, each team will have a check-in with the competition coordinators. This check-in will serve as an opportunity for teams to verify that they are on a reasonable track to complete their project in time. For this meeting, teams must prepare and submit the following documents by 03/20:

1. Initial planner
2. Current planner
3. Any available mechanical CAD assemblies and drawings

No specific format is required for the planner submission. Assemblies and parts must be submitted as .STEP files. This check-in is not part of a team's score, but is required to stay registered.

Rules:

Design:

1. Additional Components:

Teams may purchase and use additional components in their designs, as long as they do not replace the components provided in the kits. Components added can include but are not limited to sensors, ball bearings, breakout boards for sensor usage, filament, etc. Filament usage should be rounded to the nearest roll. If teams decide to include additional components in their final design, they must submit a copy of the purchase form with their final file submission before the competition. Total cost of additional components used in the final design must not exceed a value $150 per team, pre-tax. Any components team members already possess and use in the final design must also be included in this total.

Purchase form link:

HAND ERC DDC Purchase Form Template.xlsx

2. Prebuilt Solutions:
Prebuilt mechanical solutions may not be purchased or used in the final design. This includes prebuilt or open source mechanical designs. Teams are encouraged to do research on and take inspiration from existing designs, but all final mechanical designs must be the products of the team. If teams are unsure if something qualifies as an existing design, they can seek clarification from the competition coordinators to include it in their design.

3. Outside Help
Teams are permitted to consult outside help for theoretical questions, but all mechanical, electrical, and software choices must be made solely by the team. No outside assistance can be included in the final design. Teams will be expected to be able to justify their design choices, explain their design process, and document the design process.

4. Power
Teams must power their designs solely through the provided battery pack, and the USB connection to the microcontroller. When adding additional components, teams are recommended to check power requirements before purchasing.

5. Completion Date
Teams are required to complete their final design before the design presentation on 04/11, (date subject to change). Major design changes cannot be made after this point, and any significant changes will result in a disqualification.

6. Forearm Shell Usage
All components must be placed fully inside the provided forearm shell, or within the attached hand. Additionally, teams may not alter the structure of the provided forearm shell (i.e. cutting, drilling, removing material from, etc.) without explicit permission from competition coordinators. Cosmetic changes and attaching components using adhesives are permitted. Not including the forearm in the design as detailed here **will result in disqualification**.

7. AI Usage
As mentioned in rule 3, teams will be expected to understand and be able to explain and justify their design choices. We will not be restricting AI usage directly, but we highly recommend that if teams decide to use AI or LLMs in their project, they ensure that they fully understand any and all AI produced work that they integrate into their projects. Design choices and justification of design will be scored as part of the competition, so not doing so will directly impact a team's score.

8. Kit Component Usage
Teams must use all components in their kits, unless permission is explicitly given by the coordinators to do otherwise. Exceptions to this are the Spiderwire, M2&M4 nuts/bolts, and jumper wires.

9. Weight Limitations
The total design weight may not exceed 10kg. This includes the forearm, cables, electronics, hand, and everything else attached to or internal to the design. This excludes UI devices, controllers, or computers.

10. Customization
Teams may visually customize their forearms, and are encouraged to paint, draw, or otherwise write their approved team names on the provided forearms. Customizations on the outside of the forearm may not add significant volume, thickness, or durability to the forearm. Teams are expected to use their best judgement to customize, and any unprofessional or vulgar designs may result in a point loss or a disqualification at the coordinators discretion.

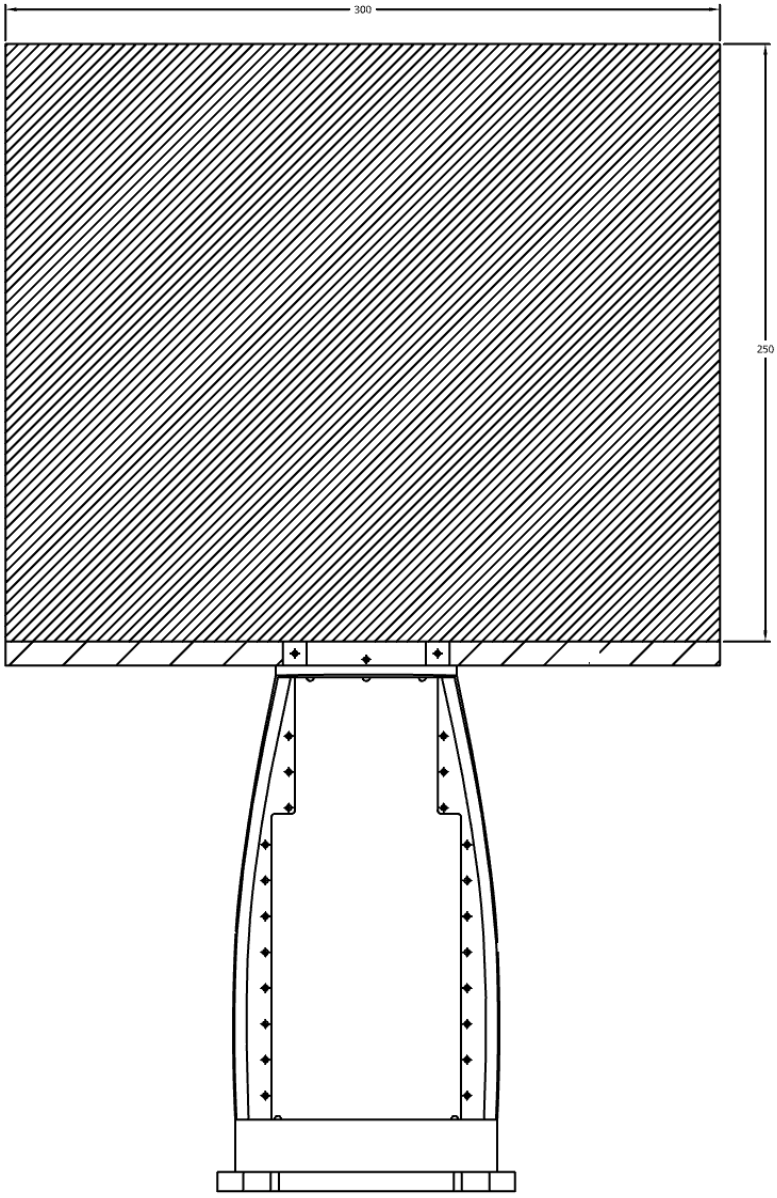
11. Degrees of Freedom and Hand Design
Designs must have a total of 11 DoFs, four fingers, one thumb, and a wrist. Each finger and the thumb must each have 1 flexion/extension DoF, and 1 splay DoF. The wrist must have 1 DoF. The wrist DoF may be flexion/extension, splay, or a rotation DoF.

On top of the DoF requirements, each finger must have three joints, and the thumb must have two. These do not need to be individually actuated, but they must be demonstrably bendable. As in, the team must be able to demonstrate

the robot's ability to bend all of the joints without physical human intervention. Not meeting this requirement may result in a point deduction per joint not bendable. (Bendable is defined in this case as >10° displacement from unpowered position for flexion/extension, or >5° for splay joints).

12. Size Limitations

Hands built must not exceed 250mm vertically from the top of the base, nor 150mm to the left or right of the center of the base. The hashed area below shows the area where the hand and wrist may be. The wide hashed area may also be occupied.



Competition Rules:

1. Design Functionality Deadline

Teams are not allowed to troubleshoot their design after their evaluation period on the competition day begins. This is non-negotiable, so teams are encouraged to thoroughly test their designs prior to the competition day.

2. Participant Interference

Unless explicitly stated otherwise in the competition guidelines, team members are not permitted to touch, remotely control, or influence their design's operation in any way during their evaluation period. *Doing so may result in a disqualification or point deduction at the coordinator's discretion.*
Any interference from one team to another team's design or work, either during the evaluation period or at any point during design, i*s subject to disqualification at the coordinator's discretion.* This includes any verbal or physical interference, regardless of intent.

3. Coordinator Interference

Competition coordinators will not assist the design's function in any way during the evaluation, nor will they assist in debugging or fixing problems.

Resources:

Design Resources:

PCA9685 Servo Driver	
Basics and Introduction	https://www.youtube.com/watch?v=y8X9X10Tn1k
Controlling 16 Servos	https://www.youtube.com/watch?v=bal2STaoQ1M

ATmega328P Nano V3.0	
Introduction Tutorial	https://www.youtube.com/watch?v=6OR7STWnIaE
Controlling 1 Servo	https://www.youtube.com/watch?v=-VYnC3NDRFs

Onshape	
Beginner Tutorial	https://www.youtube.com/watch?v=RyS3Bx0X29A
Onshape offers a self-paced course that teaches the basics of CAD. For individuals with no experience in CAD, consider attending our workshop on OnShape.	

Tendon Kinematics
Kinematics Rapid Modeling Method of Tendon-Driven Robotic Mechanisms Using the Tendon-Routing Matrix and Equivalent Radius Matrix

Misc. Competition Resources:

Forearm CAD:

HAND DDC Forearm.zip

Purchase Form:

HAND ERC DDC Additional Materials Form Template.xlsx

Information Slides:

Dexterous Design Challenge Informational.pdf

Gantt Chart Template:

HAND ERC DDC Gantt Chart Template.xlsx

Competition Guide:

Please note that all point allocations, dates, tasks, and other details here are subject to change, and not yet finalized. This section should serve as a guide on where to allocate your team's resources and time, rather than concrete goals to aim for.

Day 1 (04/11):

Teams will present their design process, choices, and results. This may be a poster or a slide presentation. Up to 40 points will be awarded to each team for critical thinking, design process, documentation, presentation, and professionalism. A more detailed rubric will be provided closer to the competition date.

Day 2 (04/12):

Day 2 of the competition will include a lineup of tasks that each team will get an opportunity to complete. Based on performance in these tasks, up to 60 points may be awarded per team, with additional points awarded for autonomous operation.

Example **Competition Lineup:**

These tasks are examples, and are not guaranteed to include any of the actual competition tasks. Point distributions may or may not match this lineup.

Task 1. Sorting objects

Teams will have 120 seconds to grasp, pickup, and release foam objects smaller than a baseball. They must either control or autonomously program their robotic hand to sort these by color, by releasing each object over the respective bin. 2 points will be awarded for each correctly sorted object, and 2 points will be deducted per incorrectly sorted or dropped object. Up to 20 points may be achieved in this task. An additional 10 points may be awarded for fully autonomous operation.

Task 2. Simon Says

Teams will have 120 seconds to have their robotic hand follow a simon says pattern of pressing keys. Four keys will light up specific colors, and make specific frequency noises. Once the pattern has completed teams will have to press the keys in the same pattern. The pattern will reset when the robotic hand presses an incorrect key. The pattern begins with one key, and every time the pattern is correctly copied, the length of the pattern will increase by one. At the end of the time period, teams will be awarded 2 points per correct key press for their longest successful pattern copied. (E.g. if a team correctly copies a 10 key long pattern, they will be awarded 20 points). Teams may earn up to 20 points for this task, with an additional 5 points for completely autonomous operation.

Task 3. ASL

Teams will have 120 seconds to use ASL to spell out a given word, for example, “TAMU”. Points will be awarded for every letter of the word correctly identifiable by the judges.

Additional Point Deductions:

Points may be deducted for the following reasons:

- -3 points per non-bendable joint, (see rule 11)
- -5 point per missing DoF, (see rule 11)
- -3 points per kit component not included without permission, (see rule 8)
- -5 points per linear cm exceeded past restrictions (see rule 12)
- -5 points per kg over weight, (weight is rounded up to nearest kg) (see rule 9)